

VIJAYAKUMAR V S

AUTOMATION & CONTROLS ENGINEER

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PROFESSIONAL SUMMARY

Automation & Controls Engineer with experience in Design, PLC/HMI development, system integration, testing, troubleshooting, and commissioning. Skilled in PLC programming using Beckhoff TwinCAT 3 and Phoenix Contact PLCnext Gen, with expertise in Structured Text, Function Blocks, UDTs, sequence control, and state-machine concepts. Experience across robotic test systems, semiconductor equipment, packaging, and textile automation, with the ability to take ownership from panel design and machine Integration through final commissioning.

TECHNICAL SKILLS

PLC Platforms: Beckhoff TwinCAT 3, Phoenix Contact PLCnext Gen

Programming: Structured Text (ST), Ladder Logic (LD), Function Block Diagram (FBD), Function Blocks, UDTs, Sequence Control, Interlocks, State-Machine Concepts

HMI / Reporting: HMI Screen Development, HTML Report Templates, JSON Variable Mapping, Automated Test-Result Reporting

Electrical Design: AutoCAD Electrical Schematics, Control Panel Design, Component Selection, BOM Preparation, Wiring, I/O Mapping, Signal Tracing

PCB & Harness: PCB Interface Coordination, PCB Netlist Creation, Mating Connector Identification, Harness Routing, Cable-Length Optimization

Motion & Drives: Elmo Drives, Panasonic Drives, Servo Systems, Drive Configuration & Validation

Communication: EtherCAT, PLC/HMI Communication, I/O Configuration, Diagnostics & Troubleshooting

Commissioning & Test: Machine Testing, Debugging, FAT/SAT, Machine Commissioning, Technical Documentation

PROFESSIONAL EXPERIENCE

ASM Technologies Ltd — Bangalore, India

Dec 2025 – Present

Engineer – Automation & Controls

► **OQC Testing Systems – Dexterity.ai , USA (3 Projects : AC/DC, Drivetrain & Harness Tester)**

- Contributed across the full lifecycle of three OQC test systems — electrical design, panel layout, component selection, BOM preparation, HMI, Function Blocks/UDTs, testing, debugging, and commissioning.
- Prepared electrical schematics and coordinated with the E-CAD team to finalize professional drawings; reviewed and submitted final drawings for approval.
- Owned the electrical interface design for the Harness Testing system: created the PCB netlist, mating-connector identification, pin/connector mapping, connector placement, and harness routing, optimizing cable lengths against machine layout.
- Coordinated with PCB and mechanical teams to resolve connector, routing, mounting, and clearance issues, including custom mounting solutions for select connectors.
- Developed HMI screens, reusable Function Blocks and UDTs, and HTML-based report templates with JSON variable mapping for automated test-result reporting.
- Performed hands-on machine testing and troubleshooting — I/O, wiring, sensor, communication, connector, harness continuity, interlock, HMI, and PCB-level checks — supporting the senior PLC engineer on sequence and program-level fixes.
- Supported Elmo drive/motion troubleshooting with suppliers and Beckhoff engineers, and took part in commissioning, FAT, documentation, and customer handover within project deadlines.

► **N2 Stoker Machine – Semiconductor Industry | Reliance Industries**

- Commissioned an Omron PLC-based system, including I/O validation, system startup, and field troubleshooting.
- Verified wiring, signals, and interlocks; calculated cable lengths and supported BOM preparation.
- Collaborated with OEM engineers during system validation and commissioning.

► **PLC Framework – AGI Milltech**

- Supported Function Block development for a Phoenix Contact PLCnext Gen-based machine, including reusable FB creation, testing, and documentation.

- ▶ **Packaging Automation System – Suryalata Spinning Mills, Hyderabad**
 - Commissioned a packaging automation system (Cone Transport System, Cone Palletizer, Auto Feeder); integrated subsystems and validated I/O, interlocks, and sequence logic during startup.
 - Supported PLC logic validation and signal debugging, and delivered on-site training to client technicians.
- ▶ **Cone Transport System (CTS) – Indo Count Industries, Maharashtra**
 - Commissioned the CTS, including I/O validation, signal tracing, and system testing.
 - Supported PLC programming and developed HMI interfaces for monitoring and control.
 - Prepared CAD drawings (electrical schematics, wiring diagrams, panel layouts), parts catalogues, and technical documentation.
- ▶ **Auto Feeder (R&D Project)**
 - Prepared CAD drawings, wiring diagrams, and BOM; contributed to PLC logic development and system integration during the R&D phase.

TECHNICAL PROJECT

- ▶ **M3SCS Conveyor System (Beckhoff TwinCAT 3)**
 - Developed conveyor control logic with state-machine design and modular automation principles, following ISA-88 concepts for equipment control.

EDUCATION

Post Graduate Diploma in Industrial Automation – SMEC Labs, 2024
Hands-on training in PLC, SCADA, HMI, and DCS with on-site panel building experience.

Bachelor of Technology (B.Tech) in Mechanical Engineering – Kerala Technological University, 2023 | CGPA: 7.3

LANGUAGES

English (Professional) · Hindi (Professional) · Malayalam (Native) · Tamil (Professional)